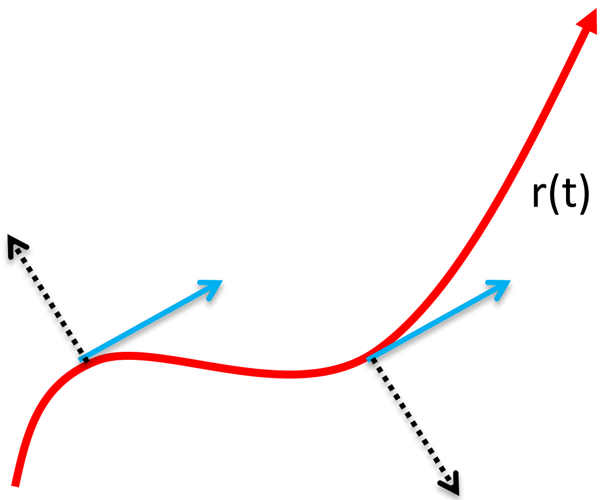
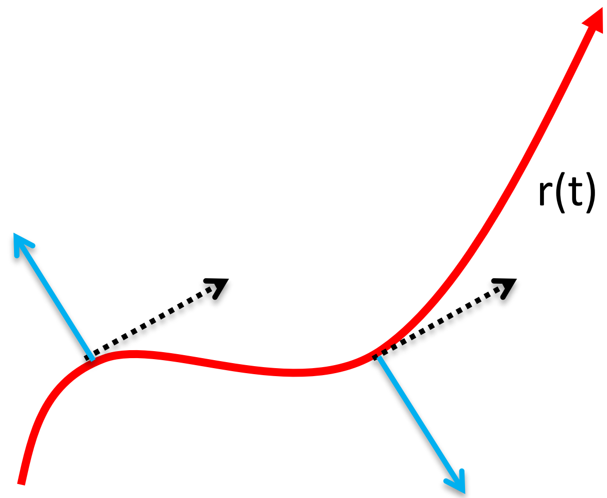
Consider a curve either in 2D or 3D



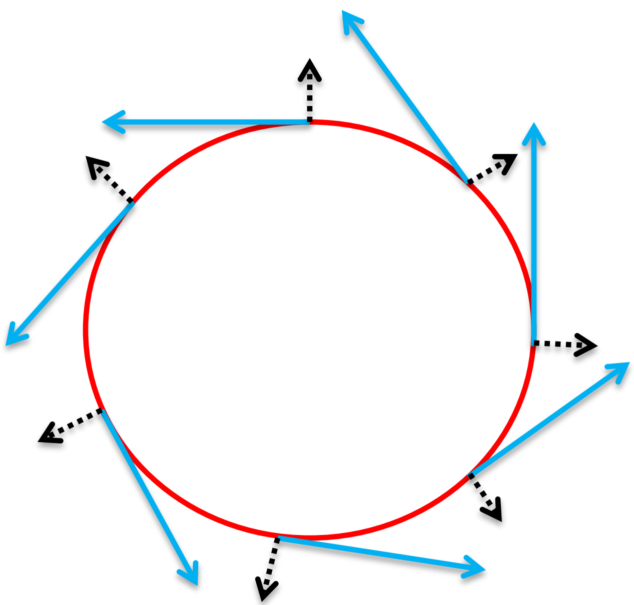
$d\mathbf{r} = \text{Tagential}$



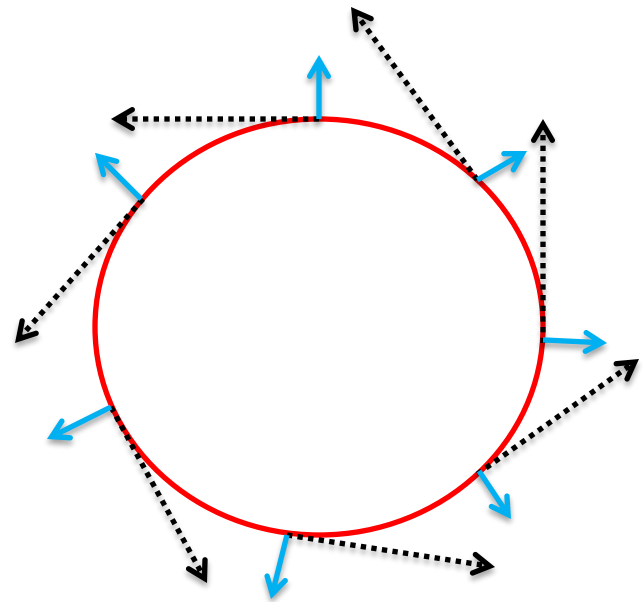
$d\mathbf{n} = |d\mathbf{r}|\mathbf{n} = \text{Radial}$

$d\mathbf{r}$ Bold Font represents Vector and dS Normal Font represents Scalar. ***

Consider a close curve (loop) either in 2D or 3D

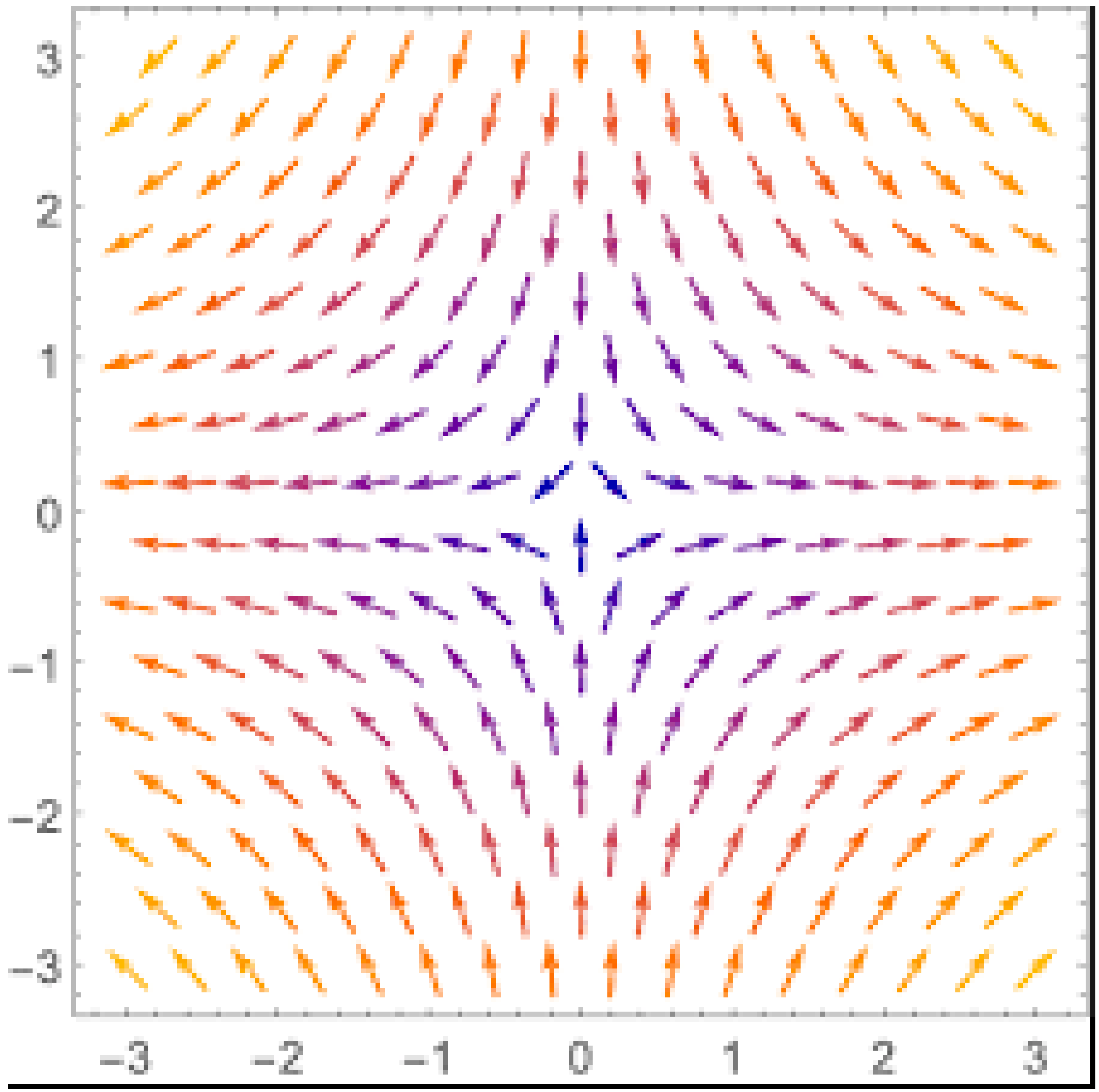


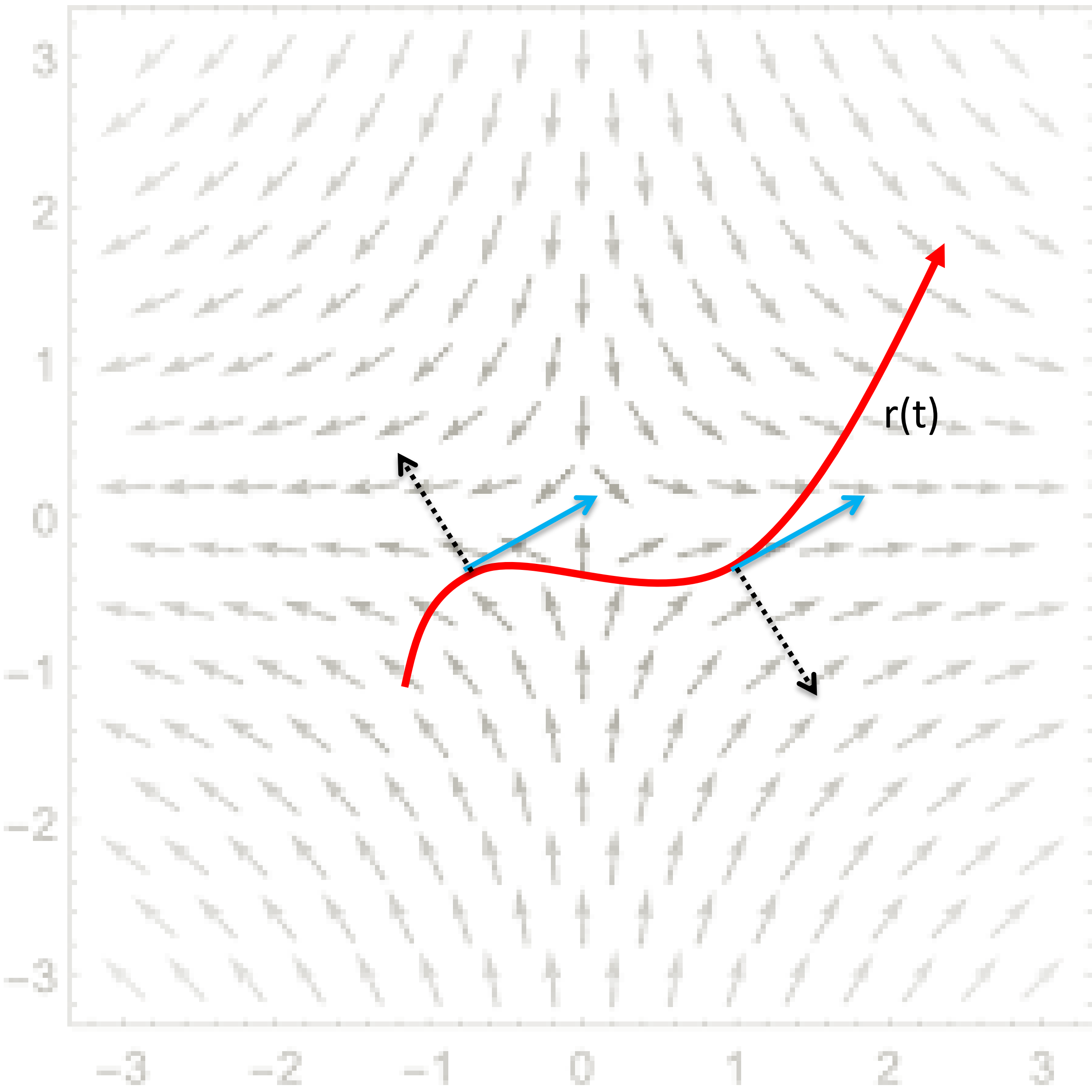
$d\mathbf{r}$ = Tagential



$d\mathbf{n} = |d\mathbf{r}|\mathbf{n}$ = Radial

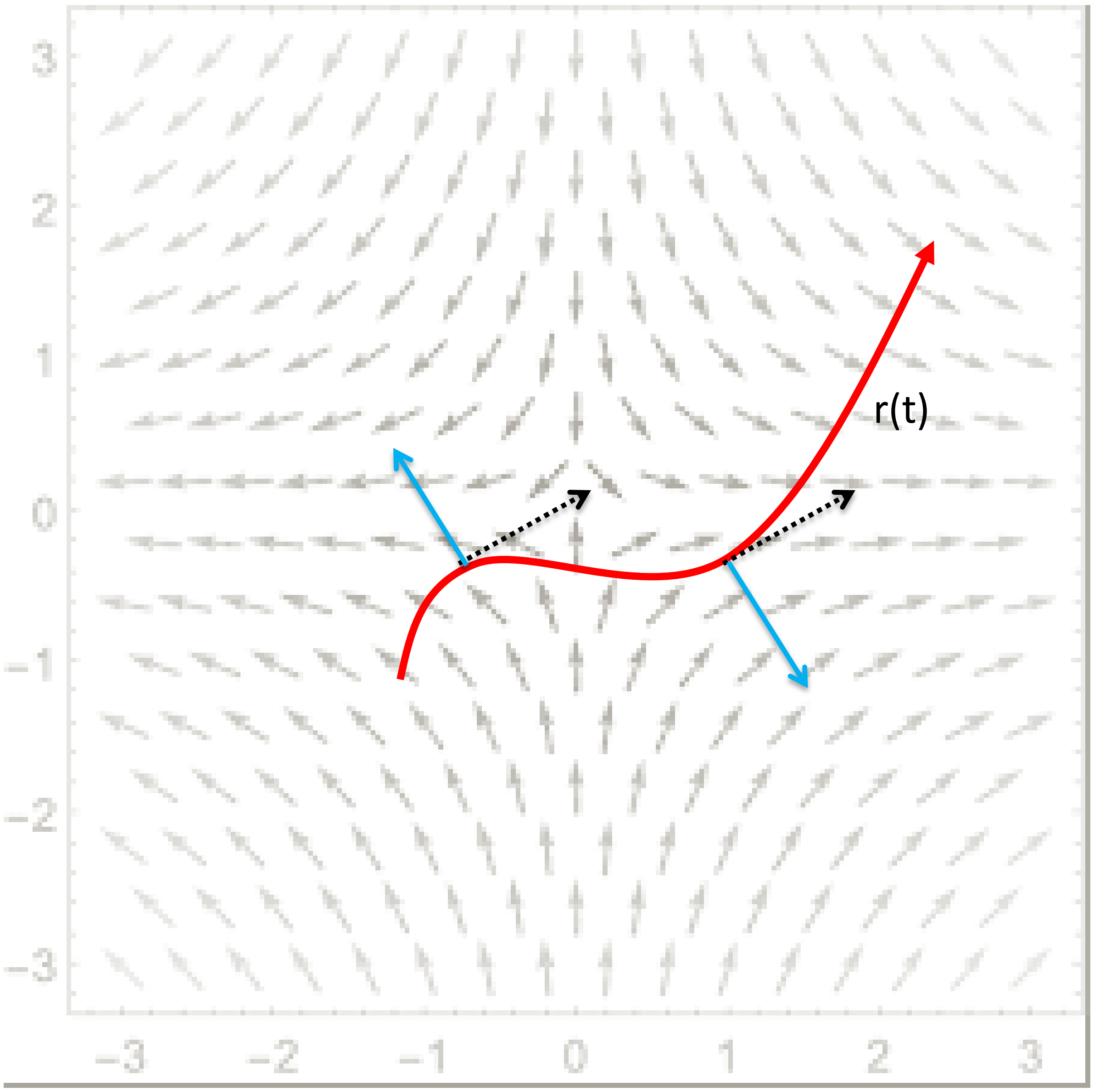
F = Vector Field





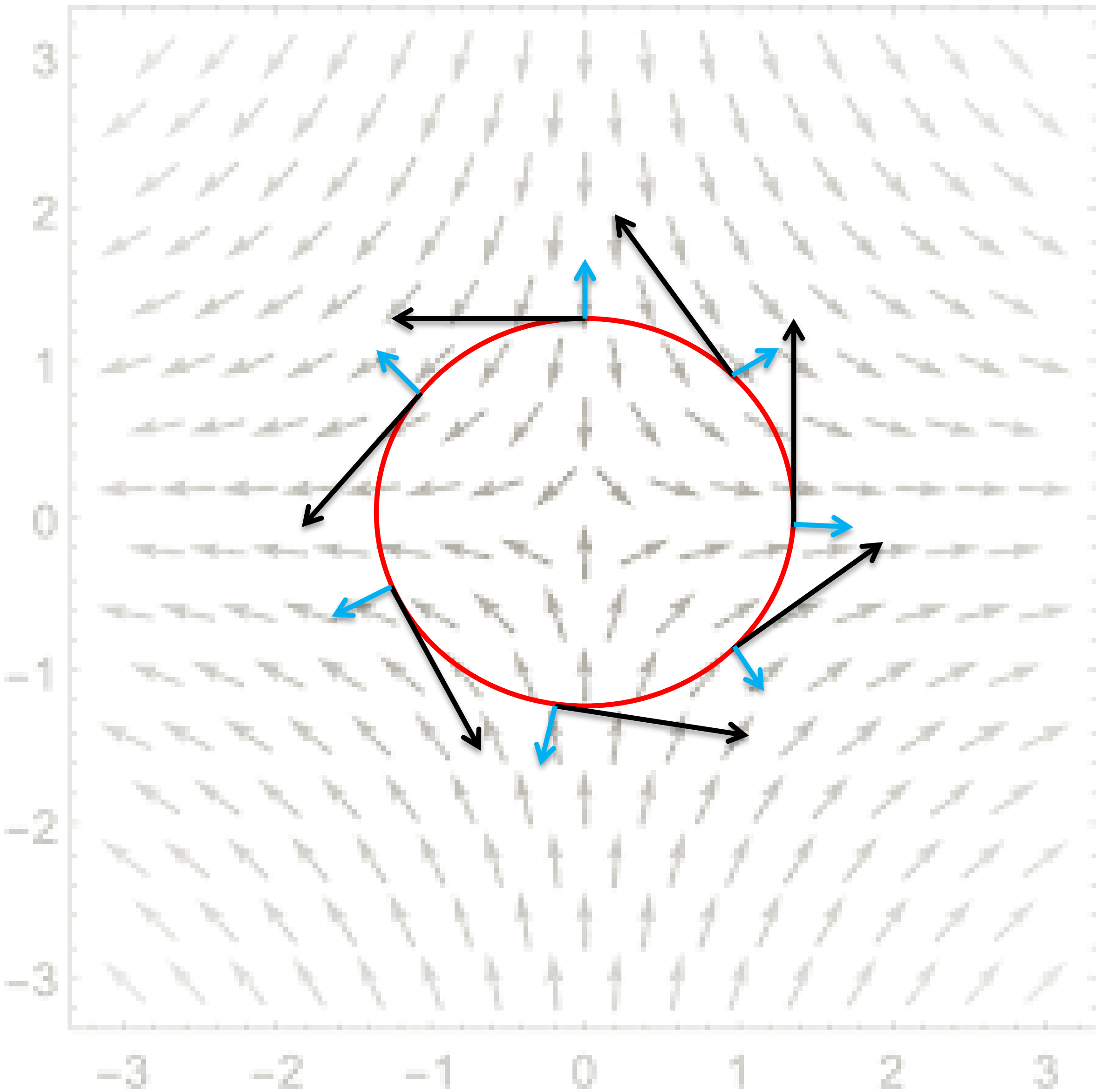
F.d**r** = dot product between **Tangential** component of the Curve and Vectors of Vector Field in that tangent direction.

$\int$**F**.d**r** = the accumulation of such dot product



F.|d**r**|**n** = dot product between **Radial** component of the curve and Vectors of Vector Field in that radial (normal) direction.

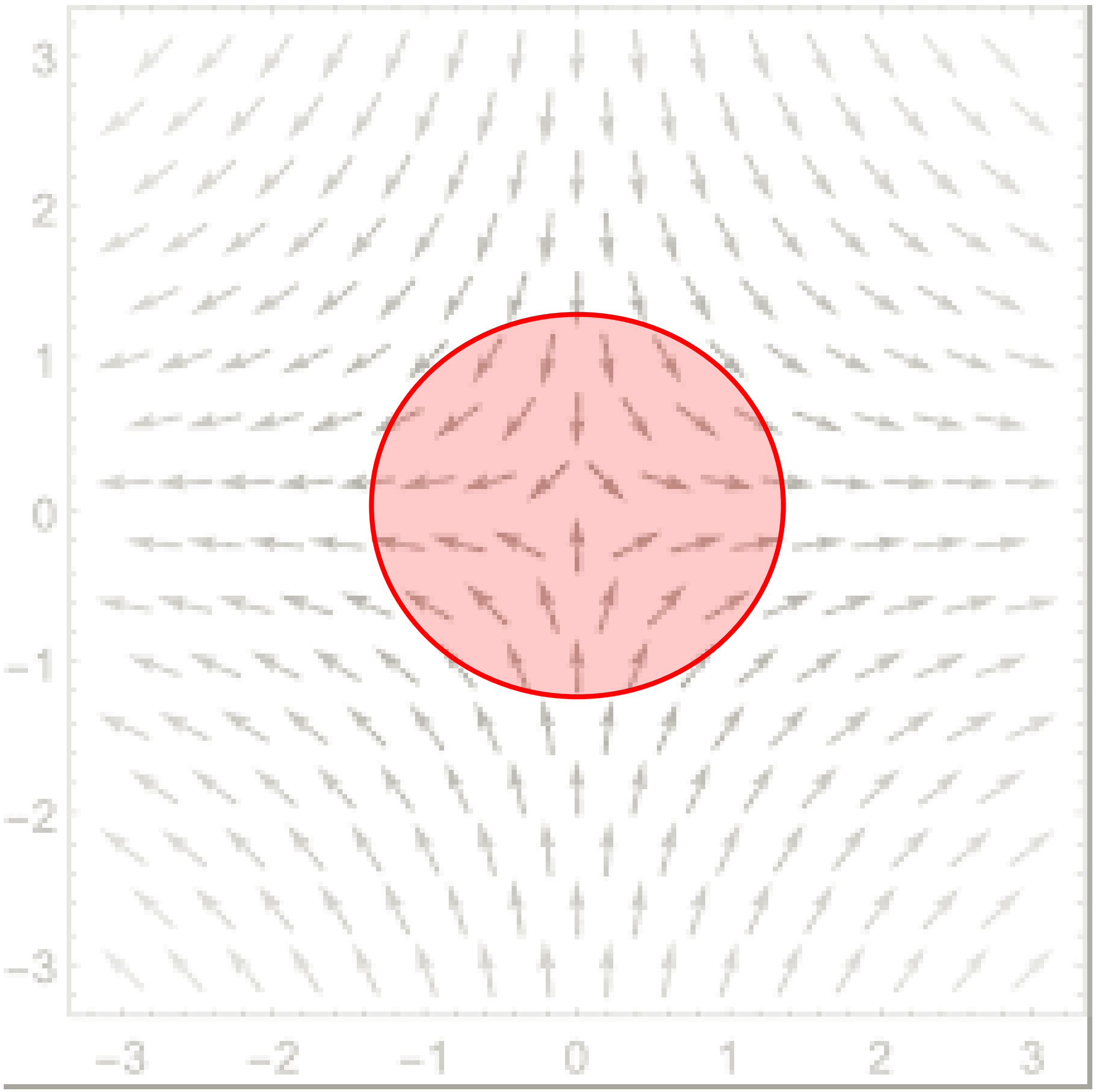
$\int$ **F**.|d**r**|**n** = is the accumulation of such dot product



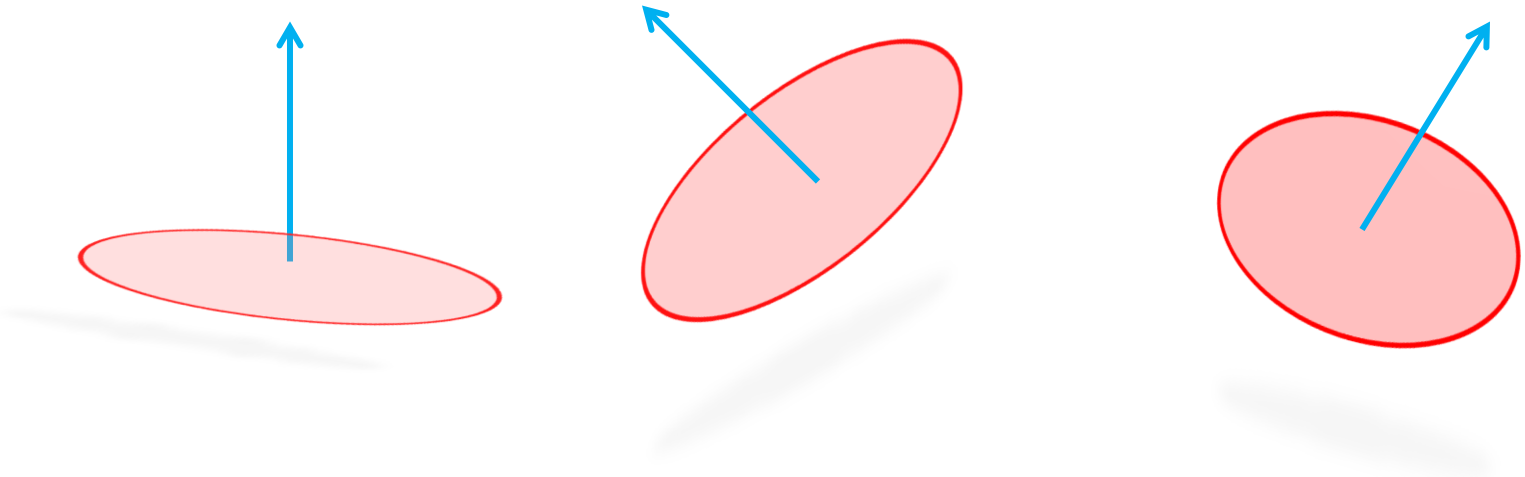
Now think about a close curve or a loop.

$\oint \mathbf{F}.d\mathbf{r} =$ Accumulation of Dot Product of Tangential Components in Vector Field

$\oint \mathbf{F}.|d\mathbf{r}|\mathbf{n} =$ Accumulation of Dot Product of Radial Components in Vector Field

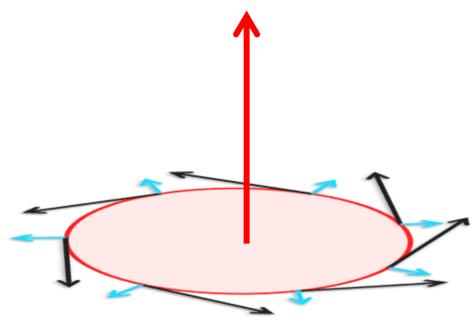


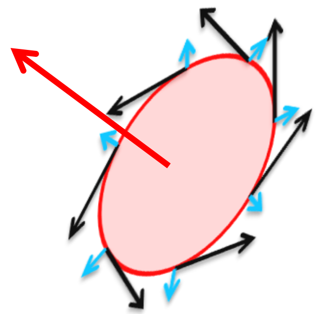
Interestingly, any close loop defines a surface bounded by its close region.



The Surface formed by a Close Curve or a Loop does not depend on 2D or 3D. A circle is a Surface in 2D or 3D. So, it does not matter.

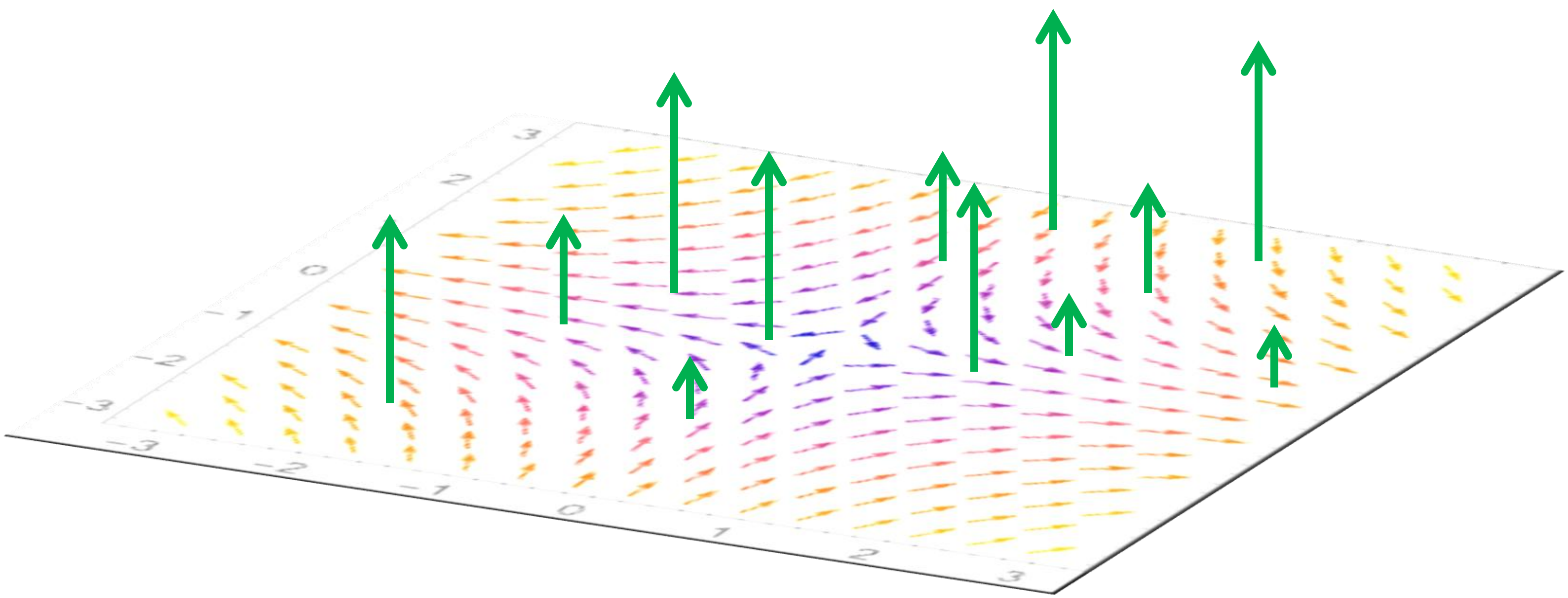
Any surface has a **Normal** called **Orientation** of the Surface.





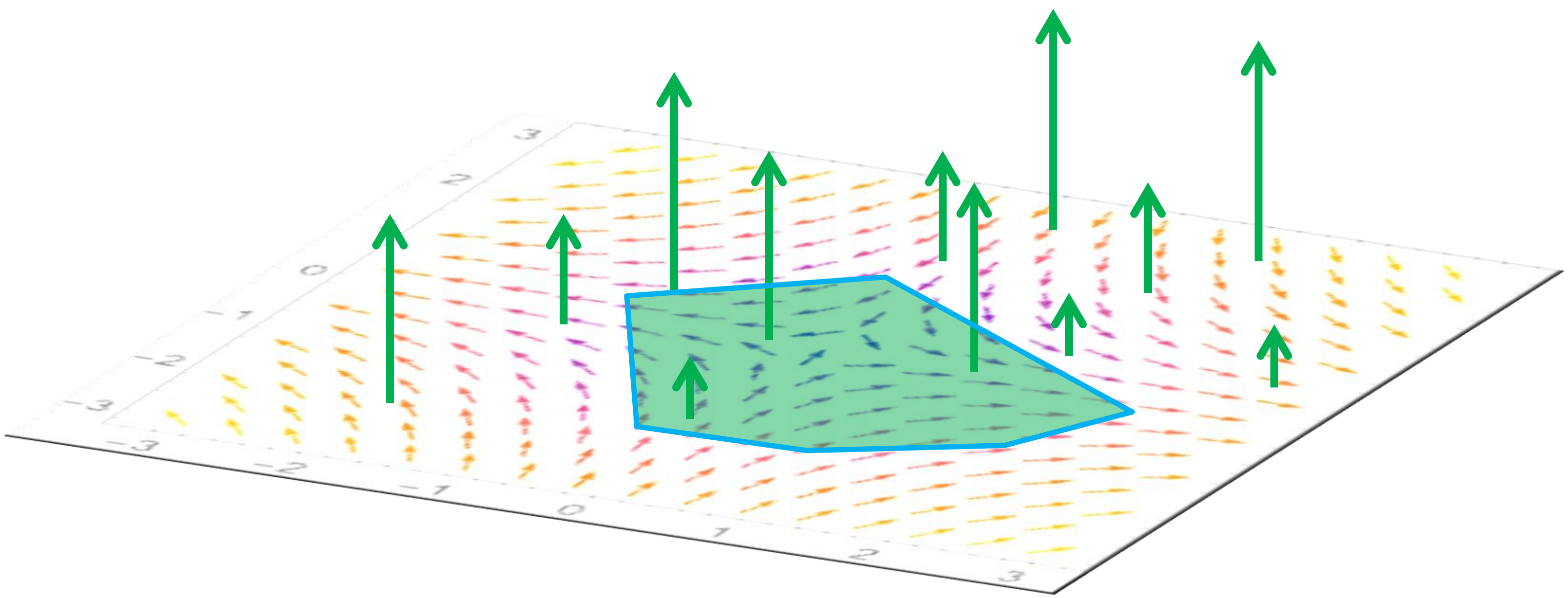
We can find the Surfaced formed by Close Loop and its Orientation (Normal) of the Surface. Of course, we will also have Tangential and Radial Components of the Close Curve or Loop as well.

$\nabla \times \mathbf{F}$



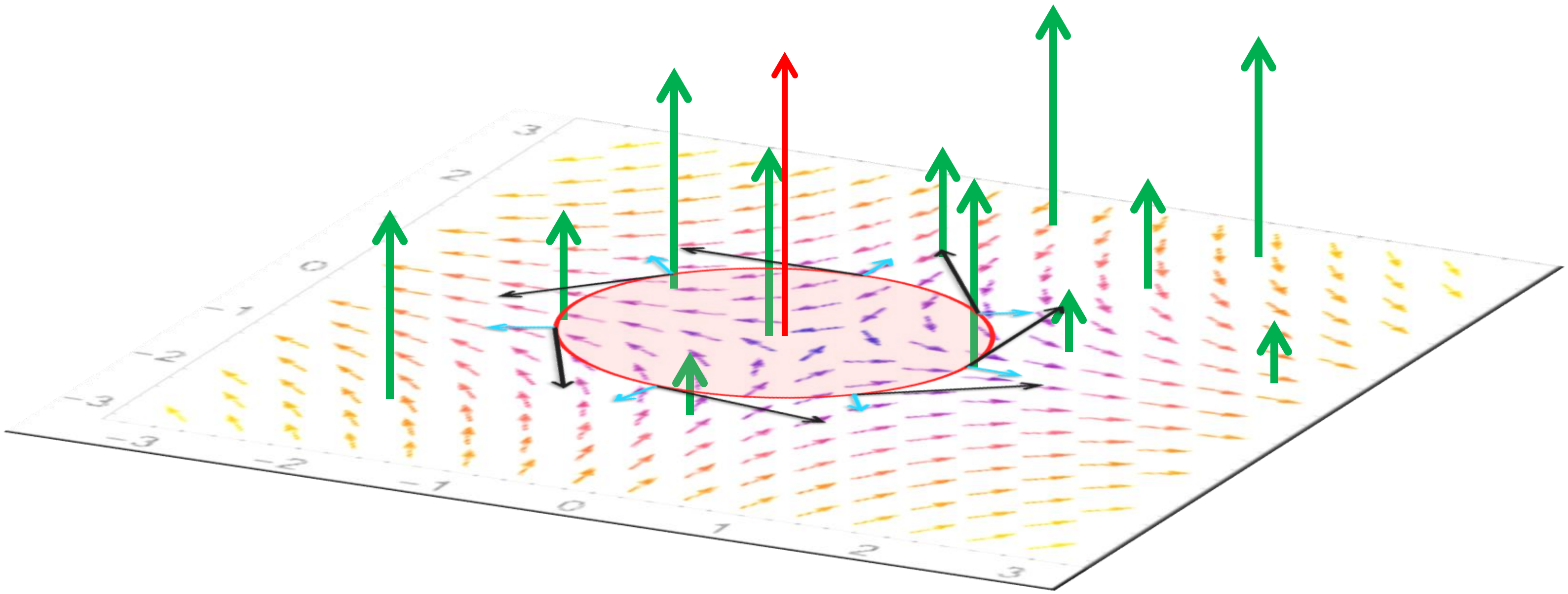
Curl of Vector Field is the Normal to the Rotation of the Vector Field.

$$\iint (\nabla \times \mathbf{F}).d\mathbf{S}$$



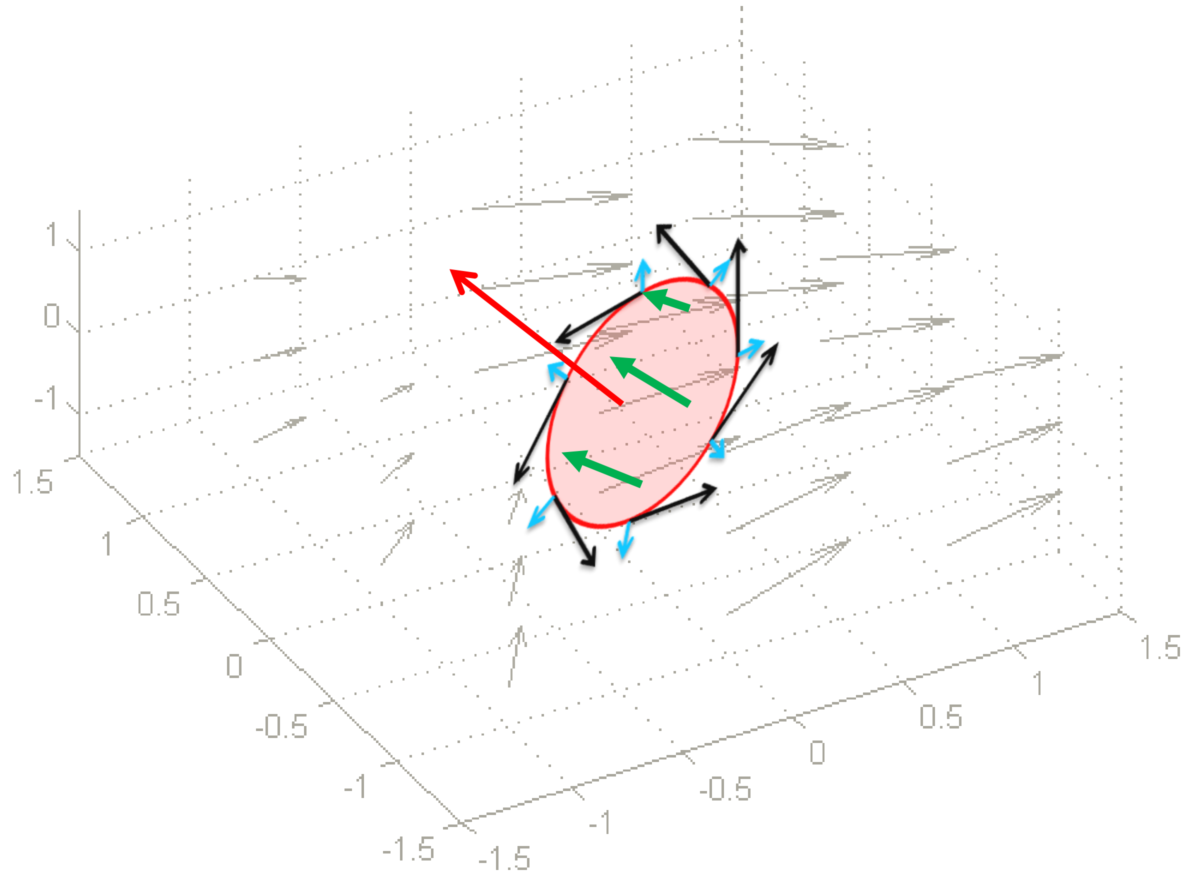
Surface Integral of **Curl** is Accumulation of Curl inside that surface

$$\oint \mathbf{F}.\mathrm{d}\mathbf{r} = \iint (\nabla \times \mathbf{F}).\mathrm{d}\mathbf{S}$$

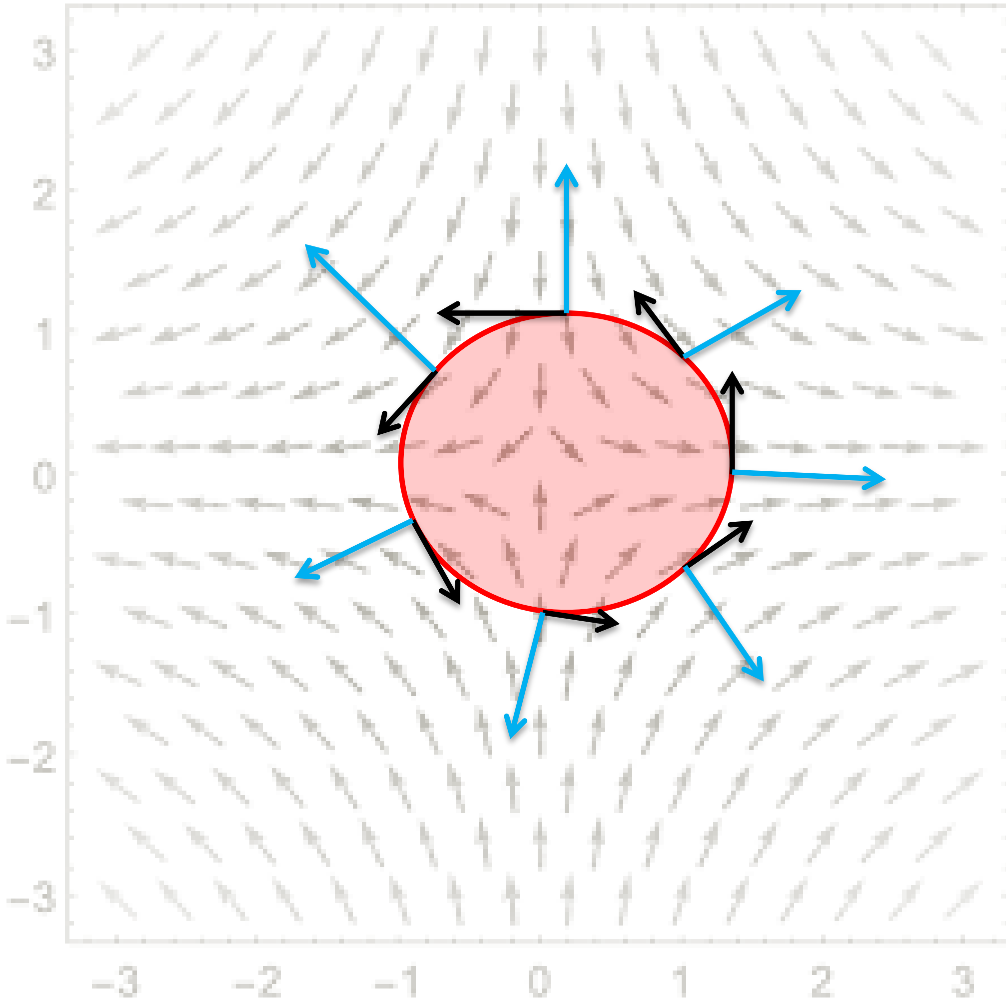


Green Theorem (1) States the Accumulation of Dot Product of **Vector Field** and **Tangential** Component of the Close Loop is equal to the Accumulation of Dot Product of **Curl** and **Orientation** of **the Surface**, defined by that Close Loop.

$$\oint \mathbf{F}.d\mathbf{r} = \iint (\nabla \times \mathbf{F}).d\mathbf{S}$$

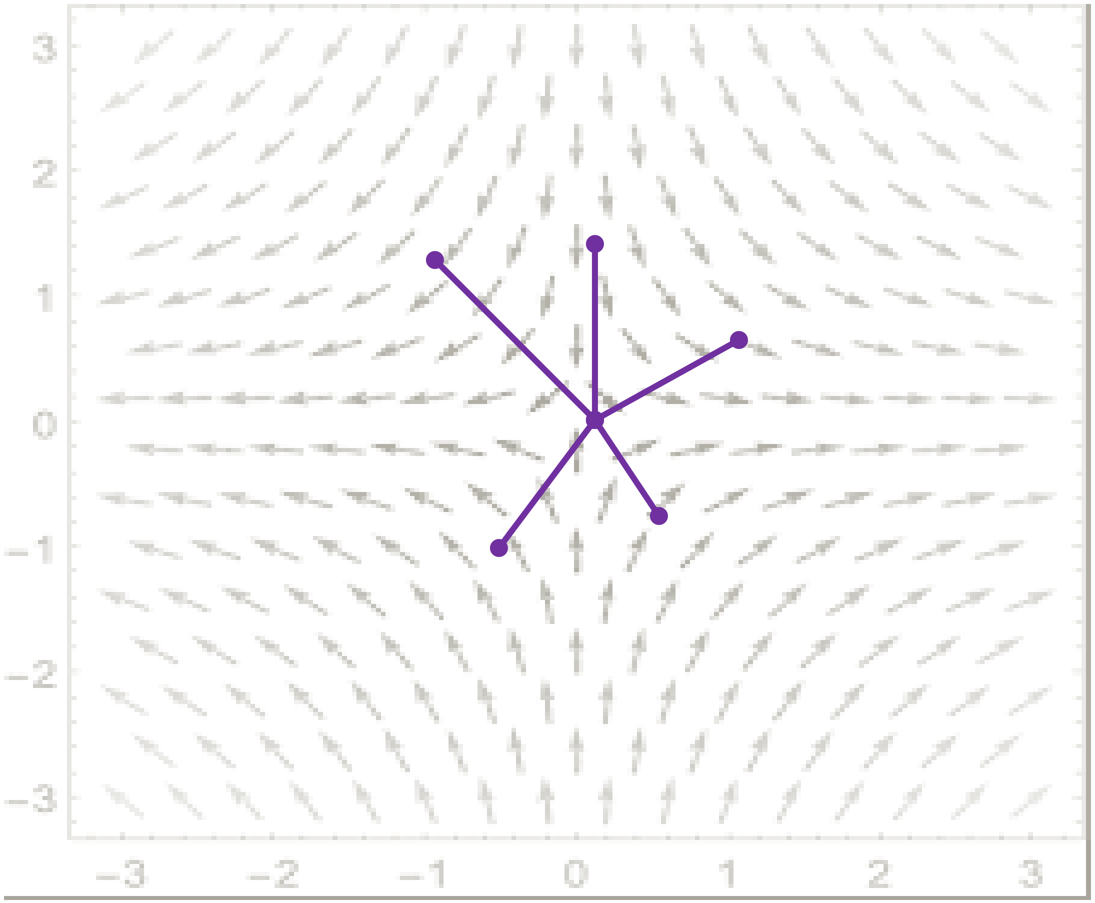


Stroke Theorem is the Extension of Green Theorem (1) to 3D, but they are the same because the Surface defined by Close Loop will always be the same either in 2D or 3D.



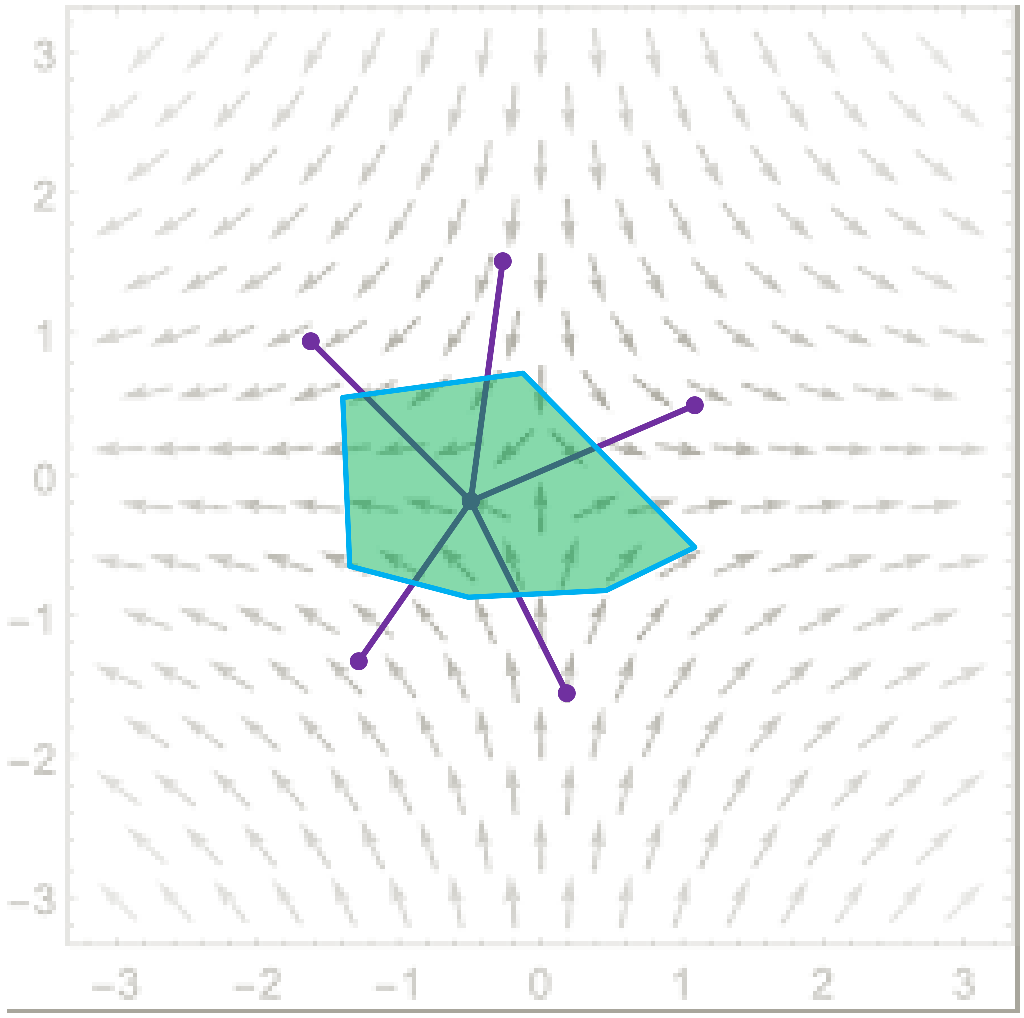
Now we are thinking about Radial Components.

$\nabla . \mathbf{F}$



Divergence of **Vector Field** is the “Scalar” measurement of “**Radiation**” of the Vector Field.

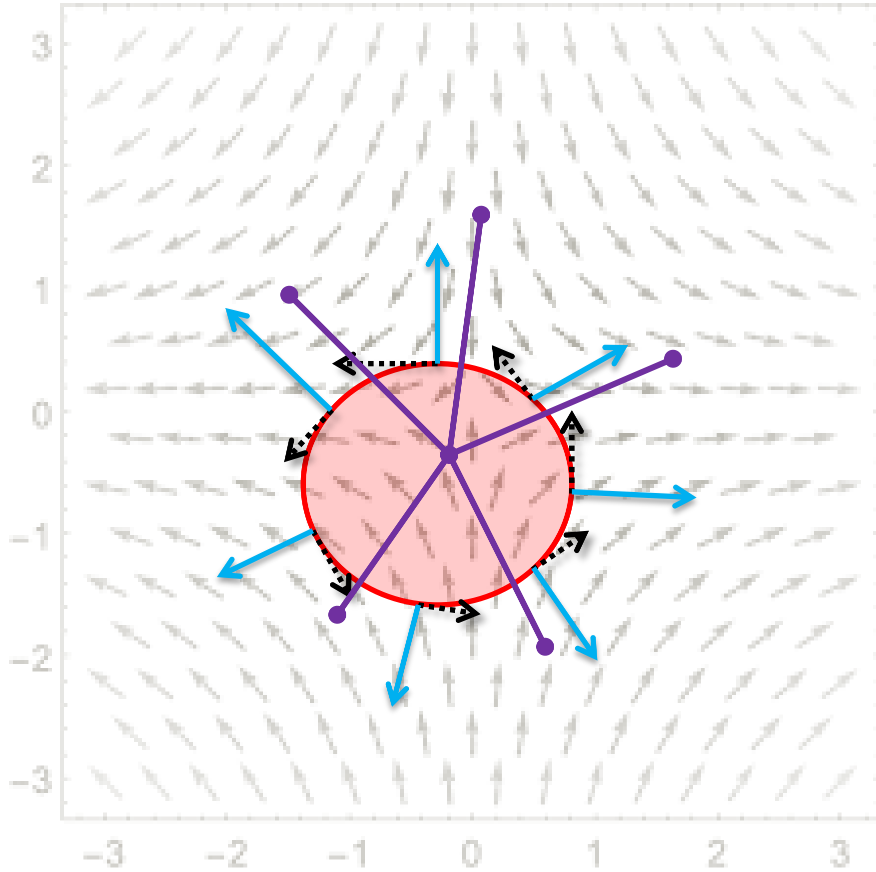
$$\iint (\nabla \cdot \mathbf{F})\, dA$$



$$dA = |d\mathbf{S}|$$

Surface Integral of **Divergence** is the Accumulation of **Radiation** inside the Area of Surface.

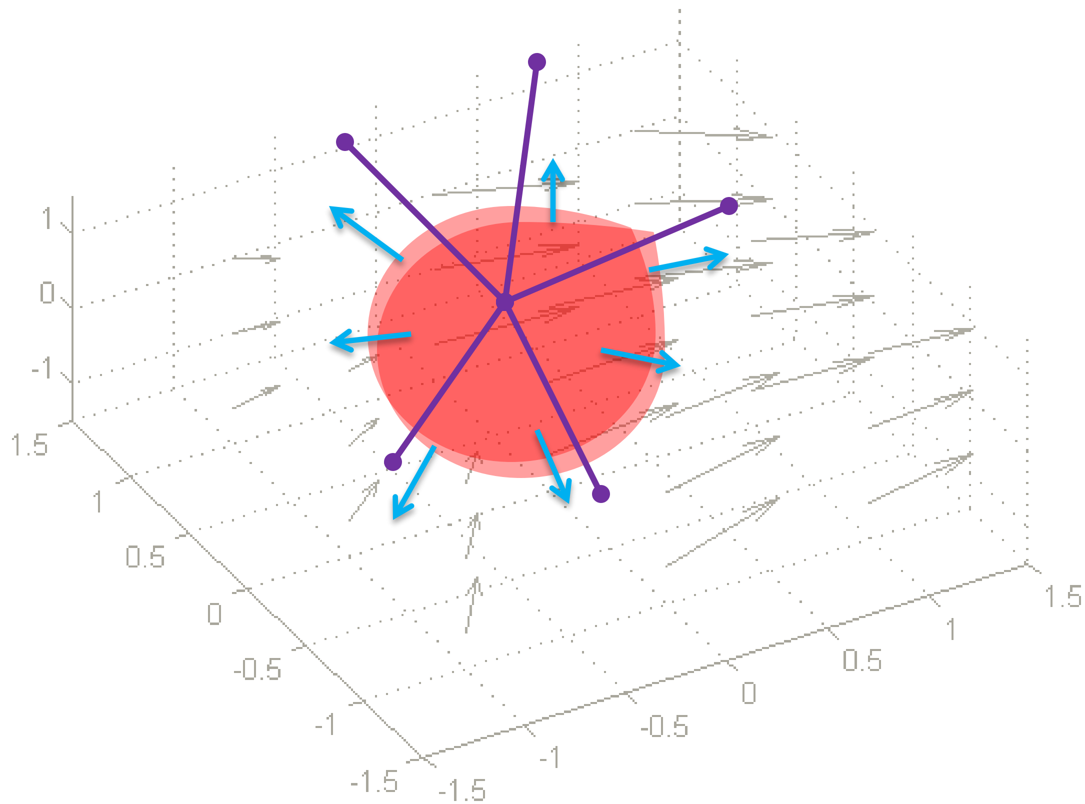
$$\oint \mathbf{F}.|d\mathbf{r}|\mathbf{n} = \iint (\nabla \, . \, \mathbf{F}) \, dA$$



$$dA = |d\mathbf{S}|$$

Green Theorem (2) States the Accumulation of Dot Product of **Vector Field** and **Radial** Component of the Close Loop is equal to the Accumulation of Product of **Divergence** and **Area of the Surface**, defined by that Close Loop.

$$\oiint \mathbf{F}.d\mathbf{S}= \iiint (\nabla . \mathbf{F})\ dV$$



Divergence Theorem states that the Accumulation of Dot Product of **Vector Field** and **Orientation** of the Close Surface is equal to the Accumulation of Product of **Divergence** and **Volume of the Close Surface.**